

Graph Cycle Detection

DFS, Backtracking, and Union-Find Methodologies

1. Understanding Cycles in Graphs

A cycle is a path that starts and ends at the same vertex without repeating any edges. Detecting cycles is a fundamental requirement in preventing deadlocks in operating systems, resolving dependencies in build systems, and ensuring network stability.

The Challenge: Cycle detection logic differs fundamentally between **Undirected** and **Directed** graphs.

2. DFS for Undirected Cycle Detection

In undirected graphs, we use DFS while keeping track of the **parent** node. A cycle exists if we encounter a visited node that is *not* the immediate parent of the current node.

```
def dfs_util(self, v, visited, parent):
    visited[v] = True
    for i in range(self.size):
        if self.adj_matrix[v][i] == 1:
            if not visited[i]:
                if self.dfs_util(i, visited, v):
                    return True
            elif parent != i: # Visited and not parent = CYCLE
                return True
    return False
```

3. DFS for Directed Cycle Detection

For directed graphs, visiting a previously visited node doesn't always imply a cycle. Instead, we must detect **Back Edges**. We use a `recursionStack` (`recStack`) to track nodes currently in the active recursion path.

Back Edge: An edge from a node to one of its ancestors in the DFS tree. If we hit a node that is already in `recStack`, a cycle is confirmed.

4. The Union-Find Algorithm (Disjoint Sets)

Union-Find is a highly efficient algorithm for undirected cycle detection. It treats each vertex as a separate set. As we process edges, we perform **Union** operations to merge sets. If both endpoints of an edge already belong to the same set (detected via **Find**), a cycle exists.

```
# Cycle detection via Find-Union
x = self.find(i)
y = self.find(j)
if x == y:
    return True # Cycle Found
self.union(x, y)
```

5. Comparison of Techniques

Algorithm	Graph Type	Main Mechanism
DFS	Both	Backtracking / RecStack
Union-Find	Undirected	Disjoint set connectivity
Kahn's (BFS)	Directed	In-degree tracking (Topological)

Case Analysis: Self-Loops and Multi-Edges (Part 1)

In real-world data, graphs often contain **self-loops** (edges starting and ending at the same node). DFS logic handles this seamlessly: if `adj_matrix[i][i] == 1`, it encounters a visited node that isn't its parent, correctly identifying a cycle.

Recursion Limit Warning: When using DFS on massive datasets (e.g., social networks), recursion depth can exceed system limits. In these cases, iterative DFS with a manual stack is the standard industry practice.

Directed Acyclic Graphs (DAGs)

A graph that passes the `is_cyclic()` check by returning `False` is known as a DAG. DAGs are essential for **Topological Sorting**, which is used to determine the order of tasks in projects where certain tasks must happen before others.

Case Analysis: Self-Loops and Multi-Edges (Part 2)

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